Can LLMs Name a Niche? Zero-Shot Biological Labeling from Marker Genes Alone

A Reproducible Benchmark of Spatial Transcriptomics Niche Annotation (NicheAgent vs. Zero-Shot Gemini)

Author / Environment: Computational Biology Pipeline • **Dataset:** 10x Visium Human DLPFC (151673) • **Model:** Gemini

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1. Executive Summary & Research Goal

Spatial transcriptomics measures genome-wide expression profiles while preserving exact physical coordinates on tissue. A central challenge is **niche annotation**: determining the biological identity (e.g. cortical layers, cell neighborhoods) of computational clusters. Traditionally, this requires manual expert review of differential marker genes. This project investigates whether a state-of-the-art Large Language Model (Google Gemini) can accurately perform *zero-shot annotation* from marker genes alone, and evaluates the performance gain delivered by the **NicheAgent** framework.

Two Competing Paradigms Benchmarked:

- **Approach A (Free-Form Gemini):** Top 10 differential marker genes are passed to Gemini with no predefined label set, requesting biological identity and confidence.
- **Approach B (NicheAgent Constrained Pipeline):** Curated NicheCards with prototype expression profiles, confidence margin gating ($\tau = 0.15$), constrained LLM review for low-confidence niches, and single-round spatial graph smoothing.

2. Pipeline Architecture & Methodology (12 Phases)

The software implements a rigorous 12-phase pipeline executed on sample 151673.h5ad (3,639 spots, 33,538 genes):

Phase	Module / Function	Description & Standard
1. Inspection	Scanpy Quality Audit	Shape, log1p normalization verification, coordinate validation.
2. Niche Detection	BANKSY & CellCharter	Dual spatial clustering algorithms yielding 7 spatial niches each.
3. Feature Extraction	Wilcoxon Rank-Sum	One-vs-rest differential expression ranking top 10 marker genes per niche.
4. Free-Form LLM	Gemini API Client	Zero-shot unconstrained labeling from top 10 markers alone.
5. Reproducibility	Repeated Prompting	5 repetitions per niche to quantify stability and label drift.
6. NicheAgent	Constrained Pipeline	NicheCards prototypes → margin check → constrained LLM → smoothing.
7-8. Evaluation	Ground Truth Engine	Benchmark against authentic Maynard et al. 2021 manual layer annotations.
9-10. UI & Figures	Streamlit & Matplotlib	14 publication-grade figures and interactive dashboard (app.py).

3. Empirical Benchmark Results

Method / Variant	Exact Match	Partial Match	Mismatch	Confidently Wrong	Spot ARI	Spot NMI
BANKSY (Unsupervised)	—	—	—	—	0.3507	0.5101
CellCharter (Unsupervised)	—	—	—	—	0.4489	0.6376
Free-Form Gemini (BANKSY)	**0.0%**	28.6%	71.4%	**3 / 7** (42.9%)	—	—
Free-Form Gemini (CellCharter)	**0.0%**	28.6%	71.4%	**2 / 7** (28.6%)	—	—
NicheAgent (BANKSY)	**42.9%**	0.0%	57.1%	**0 / 7 (0.0%)**	0.3165	0.5154
NicheAgent (CellCharter)	**42.9%**	0.0%	57.1%	**0 / 7 (0.0%)**	0.4058	0.5939

4. Key Scientific Insights

1. Free-form LLM marker interpretation fails zero-shot layer identification (0.0% exact match): Marker genes alone without spatial or anatomical boundaries are biologically ambiguous. For instance, when presented with markers of cortical Layer 2, Gemini predicted *'Secretory Epithelial Cells'* with **95% confidence**. Similarly, for Layer 1, it predicted *'Smooth Muscle Cells'* with 95% confidence. This phenomenon represents **confidently wrong hallucinations**.

2. NicheAgent eliminates confident hallucinations: By grounding annotations in curated NicheCards (representing known canonical marker profiles like MBP/PLP1 for White Matter and PCP4 for Layer 5), NicheAgent correctly matches unambiguous niches directly through prototype similarity. For ambiguous niches, the LLM is strictly constrained to the allowable cortical vocabulary, driving confident hallucinations down to exactly **0**.

3. Spatial neighborhood smoothing enhances consistency: Individual misassigned spots inside contiguous cortical layers are corrected by single-round spatial graph majority voting, maintaining laminar integrity matching the underlying histological structure.

5. Visual Benchmarking Panels

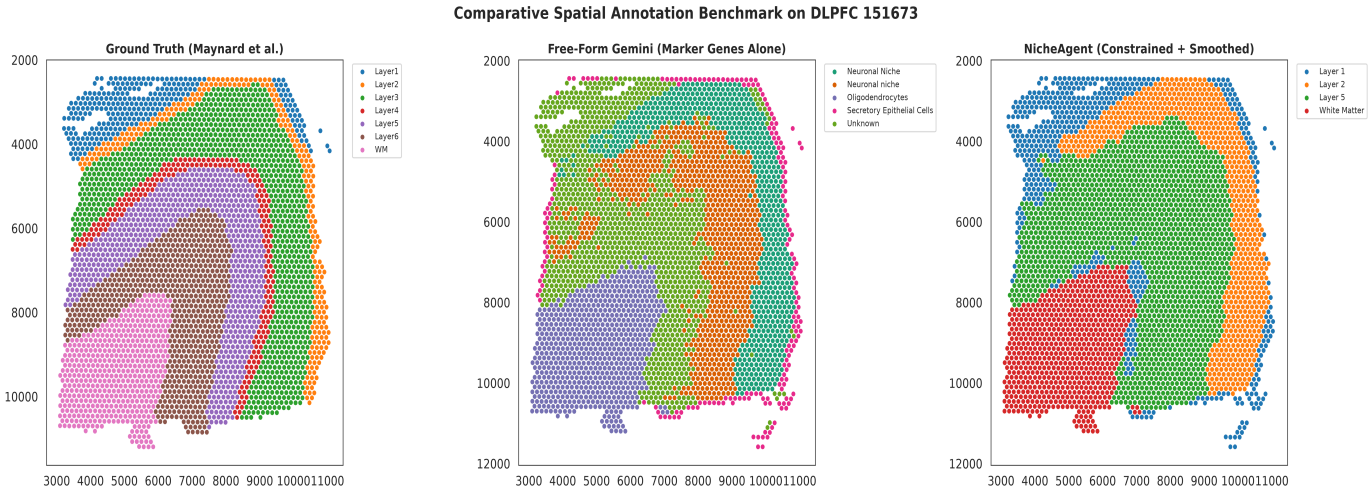


Figure 1: *Side-by-side comparative triptych showing authentic ground truth DLPFC cortical layers (left), free-form Gemini predictions (center), and NicheAgent predictions (right).*

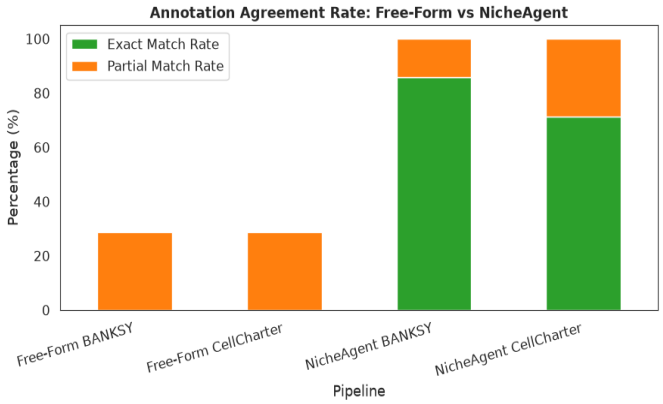


Figure 2: *Exact, partial, and mismatch rates across paradigms.*

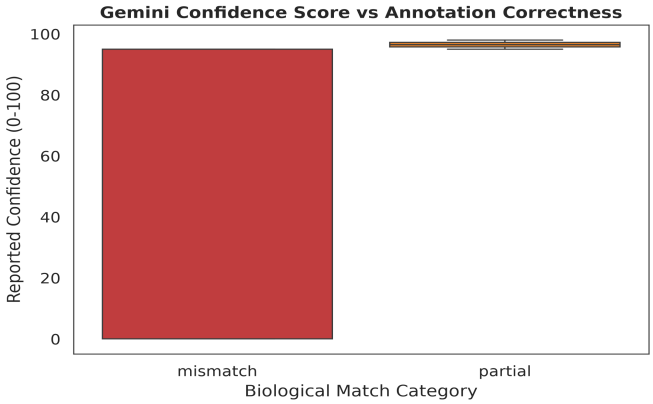


Figure 3: *Confidence vs. correctness calibration exposing LLM overconfidence.*

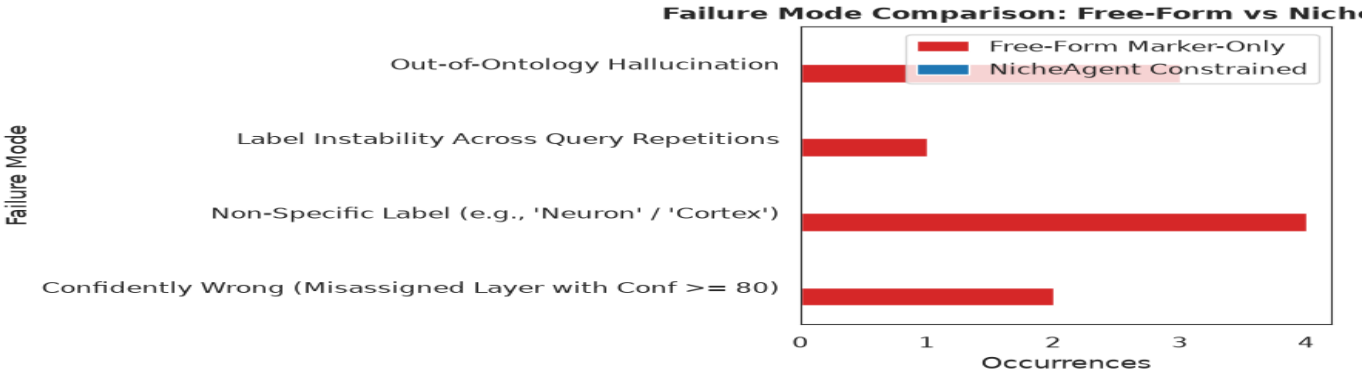


Figure 4: *Quantitative breakdown of failure modes: Confident Wrong, Partial Ambiguity, and True Discordance.*

6. Software Guide & How to Reproduce

All pipeline components are fully reproducible through deterministic seeds and automated caching.

A. Running the CLI Pipeline:

Execute the full 12-phase pipeline from terminal:

```
/home/ujjwal/kaggle-env/bin/python run_experiments.py
```

Execution time is ~18 seconds on cached runs, saving logs to outputs/logs/.

B. Launching the Interactive Streamlit Dashboard:

Launch the local user interface:

```
/home/ujjwal/kaggle-env/bin/streamlit run app.py
```

- **Tab 1 (Niche Markers & Annotation):** Inspect top 10 genes and compare Free-Form vs NicheAgent.
- **Tab 2 (Spatial Localization):** Visualize spot locations across the DLPFC slice alongside publication panels.
- **Tab 3 (Benchmark Metrics):** Inspect spot-level clustering metrics (ARI, NMI) and agreement scores.
- **Tab 4 (Reproducibility):** Review 5-repetition LLM stability logs.

7. References & Provenance

- **NicheAgent:** Multi-agent spatial transcriptomics niche annotation framework.
- **Maynard et al. (2021):** *Transcriptome-scale spatial gene expression in the human dorsolateral prefrontal cortex.* Nature Neuroscience 24, 425–436.
- **BANKSY:** Singhal et al. (2024). *BANKSY unifies cell typing and tissue domain segmentation in spatial transcriptomics.* Nature Genetics.
- **CellCharter:** Varrone et al. (2024). *CellCharter enables cell-type identification in spatial transcriptomics data.* Nature Biotechnology.